

Week -7 PRATICAL

WEEK:7

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings

```
#include <stdio.h>
#include <stdlib.h>
int global_var = 100;
static int static_var = 200;
void code_function()
{
    printf("Inside code_function()\n");
}
int main()
{
    int local_var = 10;
    int *heap_var = malloc(sizeof(int));
    if (heap_var == NULL)
    {
        printf("Memory allocation failed\n");
        return 1;
    }
    *heap_var = 300;
    printf("\n===== MEMORY ADDRESS DEMONSTRATION =====\n\n");
    printf("Code segment:\n");
    printf("Address of code_function : %p\n",
        (void *)code_function);
    printf("\nGlobal segment:\n");
```

```

printf("Address of global_var : %p\n",
(void *)&global_var);

printf("\nStatic segment:\n");

printf("Address of static_var : %p\n",
(void *)&static_var);


printf("\nHeap segment:\n");

printf("Address of heap_var : %p\n",
(void *)heap_var);

printf("\nStack segment:\n");

printf("Address of local_var : %p\n",
(void *)&local_var);

printf("\n===== \n");

free(heap_var);

return 0;

}

```

```

yaswanth6970@Yash6970:~/signal_project$ cd ~
mkdir -p ~/week7
cd ~/week7
yaswanth6970@Yash6970:~/week7$ pwd
/home/yaswanth6970/week7
yaswanth6970@Yash6970:~/week7$ nano addresses.c
yaswanth6970@Yash6970:~/week7$ gcc -Wall -Wextra addresses.c -o addresses
yaswanth6970@Yash6970:~/week7$ ls
addresses  addresses.c
yaswanth6970@Yash6970:~/week7$ ./addresses

===== MEMORY ADDRESS DEMONSTRATION =====

Code segment:
Address of code_function : 0x5c716dde71c9

Global segment:
Address of global_var : 0x5c716ddea010

Static segment:
Address of static_var : 0x5c716ddea014

Heap segment:
Address of heap_var : 0x5c7171946010

Stack segment:
Address of local_var : 0x7ffe862baf0c

=====

```

Analyze Linux Process Address Space

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;
static int static_var = 200;

int main()
{
    int stack_var = 300;

    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        perror("malloc");
        return 1;
    }

    *heap_var = 400;

    printf("Process ID: %d\n", getpid());

    printf("Address of code function: %p\n", (void *)main);
    printf("Address of global_var : %p\n", (void *)&global_var);
    printf("Address of static_var : %p\n", (void *)&static_var);
    printf("Address of heap_var : %p\n", (void *)heap_var);
    printf("Address of stack_var : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");
    printf("Press ENTER to terminate.\n");

    getchar();

    free(heap_var);

    return 0;
```

```
yaswanth6970@Yash6970:~/week7$ nano memory_layout.c
yaswanth6970@Yash6970:~/week7$
yaswanth6970@Yash6970:~/week7$ gcc -Wall -Wextra memory_layout.c -o memory_layout
yaswanth6970@Yash6970:~/week7$ ./memory_layout
Process ID: 1178
Address of code function: 0x5a05bf476229
Address of global_var : 0x5a05bf479010
Address of static_var : 0x5a05bf479014
Address of heap_var : 0x5a05c2112010
Address of stack_var : 0x7ffe453ed0fc

Process is running...
Press ENTER to terminate.
|
}
```

/proc/<PID>/maps

```
yaswanth6970@Yash6970: ~/1 × + ▾
yaswanth6970@Yash6970:/mnt/c/WINDOWS/system32$ cd ~/week7
yaswanth6970@Yash6970:~/week7$ cat /proc/1178/maps
5a05bf475000-5a05bf476000 r--p 00000000 08:30 40705 /home/yaswanth6970/week7/memory_layout
5a05bf476000-5a05bf477000 r-xp 00001000 08:30 40705 /home/yaswanth6970/week7/memory_layout
5a05bf477000-5a05bf478000 r--p 00002000 08:30 40705 /home/yaswanth6970/week7/memory_layout
5a05bf478000-5a05bf479000 r--p 00002000 08:30 40705 /home/yaswanth6970/week7/memory_layout
5a05bf479000-5a05bf47a000 rw-p 00003000 08:30 40705 /home/yaswanth6970/week7/memory_layout
5a05c2112000-5a05c2133000 rw-p 00000000 00:00 0 [heap]
7e2dd8800000-7e2dd8828000 r--p 00000000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7e2dd8828000-7e2dd89c0000 r-xp 00028000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7e2dd89c0000-7e2dd8a0e000 r--p 001c0000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7e2dd8a0e000-7e2dd8a12000 r--p 0020d000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7e2dd8a12000-7e2dd8a14000 rw-p 00211000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7e2dd8a14000-7e2dd8a21000 rw-p 00000000 00:00 0
7e2dd8bf3000-7e2dd8bf6000 rw-p 00000000 00:00 0
7e2dd8bf6000-7e2dd8bfd000 rw-p 00000000 00:00 0
7e2dd8bfd000-7e2dd8c01000 r--p 00000000 00:00 0 [vvar]
7e2dd8c01000-7e2dd8c03000 r--p 00000000 00:00 0 [vvar_vclock]
7e2dd8c03000-7e2dd8c05000 r-xp 00000000 00:00 0 [vdso]
7e2dd8c05000-7e2dd8c06000 r--p 00000000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2
7e2dd8c06000-7e2dd8c35000 r-xp 00001000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2
7e2dd8c35000-7e2dd8c40000 r--p 00030000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2
7e2dd8c40000-7e2dd8c42000 r--p 0003b000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2
7e2dd8c42000-7e2dd8c43000 rw-p 0003d000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2
7e2dd8c43000-7e2dd8c44000 rw-p 00000000 00:00 0
7ffe453cf000-7ffe453f0000 rw-p 00000000 00:00 0 [stack]

yaswanth6970@Yash6970:~/week7$ cat /proc/1178/maps | grep heap
5a05c2112000-5a05c2133000 rw-p 00000000 00:00 0 [heap]
yaswanth6970@Yash6970:~/week7$ cat /proc/1178/maps | grep stack
7ffe453cf000-7ffe453f0000 rw-p 00000000 00:00 0 [stack]
yaswanth6970@Yash6970:~/week7$ cat /proc/1178/maps | grep memory_layout
5a05bf475000-5a05bf476000 r--p 00000000 08:30 40705 /home/yaswanth6970/week7/memory_layout
5a05bf476000-5a05bf477000 r-xp 00001000 08:30 40705 /home/yaswanth6970/week7/memory_layout
5a05bf477000-5a05bf478000 r--p 00002000 08:30 40705 /home/yaswanth6970/week7/memory_layout
5a05bf478000-5a05bf479000 r--p 00002000 08:30 40705 /home/yaswanth6970/week7/memory_layout
5a05bf479000-5a05bf47a000 rw-p 00003000 08:30 40705 /home/yaswanth6970/week7/memory_layout
yaswanth6970@Yash6970:~/week7$ cat /proc/1178/maps | grep -E '\[heap\]|\[stack\]'
5a05c2112000-5a05c2133000 rw-p 00000000 00:00 0 [heap]
7ffe453cf000-7ffe453f0000 rw-p 00000000 00:00 0 [stack]
yaswanth6970@Yash6970:~/week7$ cat -n /proc/1178/maps
1 5a05bf475000-5a05bf476000 r--p 00000000 08:30 40705 /home/yaswanth6970/week7/memory_l
ayout
2 5a05bf476000-5a05bf477000 r-xp 00001000 08:30 40705 /home/yaswanth6970/week7/memory_l
ayout
3 5a05bf477000-5a05bf478000 r--p 00002000 08:30 40705 /home/yaswanth6970/week7/memory_l
ayout
4 5a05bf478000-5a05bf479000 r--p 00002000 08:30 40705 /home/yaswanth6970/week7/memory_l
ayout
5 5a05bf479000-5a05bf47a000 rw-p 00003000 08:30 40705 /home/yaswanth6970/week7/memory_l
ayout
6 5a05c2112000-5a05c2133000 rw-p 00000000 00:00 0 [heap]
7 7e2dd8800000-7e2dd8828000 r--p 00000000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so
.6
8 7e2dd8828000-7e2dd89c0000 r-xp 00028000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so
.6
9 7e2dd89c0000-7e2dd8a0e000 r--p 001c0000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so
.6
10 7e2dd8a0e000-7e2dd8a12000 r--p 0020d000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so
.6
11 7e2dd8a12000-7e2dd8a14000 rw-p 00211000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so
.6
12 7e2dd8a14000-7e2dd8a21000 rw-p 00000000 00:00 0
```